Chapter 38
Value at Risk and Expected Shortfall

Earlier chapters took one instrument or one market at a time: how it works, who uses it, and what it does to the risks of the people who hold it. This one asks what a whole portfolio of positions can lose. That is a different question, and its honest answer is a distribution of losses rather than one figure.

The difficulty is not computational. A trading book's loss distribution can be simulated to any precision the machine allows. The difficulty is that a risk report, a capital rule, a limit framework, and a margin call each need a single number. Everything in this chapter is a way of choosing which number, and the choices are not equivalent—one of them, the one the industry used for twenty years, can penalize diversification.

Two things make this chapter necessary rather than optional. Chapter 39 sets capital from these measures, so an error here is an error in what a bank must hold. Expected shortfall is also a standard objective in deep hedging, the learned hedging strategies of the online note [TN41], so a reader who does not have it defined cannot read what that work optimizes.

Sections 38.1 to 38.4 define the two measures and the property that separates them. Sections 38.5 to 38.9 compute them by historical simulation, by a model, for books with options, under fat tails, and for losses not yet observed; Sections 38.10 to 38.14 cover the inputs: volatility, correlation, copulas, factor reduction, and scaling across horizons. Sections 38.15 to 38.18 put the measures to use: attribution to positions, backtesting, stress testing, and stressed calibration. Section 38.19 says what the measures cannot see, Section 38.20 who uses them and why, and Section 38.21 where the ideas came from.

38.1 WHAT A RISK MEASURE IS FOR

A **risk measure** maps a loss distribution to a single number. Three uses pull in different directions and it is worth separating them before choosing.

A limit is a control. It must be stable enough that a trader can plan against it and responsive enough to bind when it should, and, compared across desks, it must add up.

Capital is a buffer. It answers "how much loss must be survivable," so it cares about the size of bad outcomes, not their frequency.

A margin requirement is a forecast over a specific horizon, the margin period of risk of Section 8.4, which is set by how long a close-out takes rather than by anyone's risk appetite.

These are three different questions and the market answers them with variants of one idea, a number read off the tail of the loss distribution. A measure that appears to behave strangely is often being asked a question it was not built for.

38.2 VALUE AT RISK

Value at risk (VaR) at confidence c over horizon h is the loss that is exceeded with probability $1 - c$:

$$\boxed{\mathbb{P}\big(L > \mathrm{VaR}_c\big) = 1 - c}, \tag{38.1}$$

where $L = V_0 - V_h = -\Delta V$ is the portfolio's loss over the horizon h, its value now less its value at the horizon's end, positive when value is lost.

It is a quantile, and the whole of its behavior follows from that. It says where the tail begins and nothing whatever about what is in it.

Where the loss distribution jumps, as a bond portfolio's does at default, or is a finite sample, (38.1) can have no solution or many. VaR is then the smallest loss l with $\mathbb{P}(L \le l) \ge c$, the lower quantile, which is how Examples 38.2 and 38.3 read it.

Its appeal was never statistical. A single currency amount, attached to a stated probability, can be put in front of a board. Section 38.20 argues that this is the reason it displaced everything that preceded it, not any property in (38.1).

38.3 EXPECTED SHORTFALL

Expected shortfall (ES) at confidence c is the average loss over the worst $1 - c$ of outcomes. Each level p beyond c has its own VaR, and expected shortfall is their average:

$$\boxed{\mathrm{ES}_c = \frac{1}{1-c}\int_c^1 \mathrm{VaR}_p \, dp}. \tag{38.2}$$

Because it averages *all* the VaRs beyond c, it reads the whole tail rather than one point on it.

When the loss distribution is continuous, (38.2) is the mean loss given that the VaR is exceeded, $\mathbb{E}[L \mid L > \mathrm{VaR}_c]$, a quantity called the conditional tail expectation. Where the distribution jumps at the VaR, as in Example 38.2, the two differ, and (38.2) is the definition, because it is the form that keeps the measure coherent. The same quantity goes by other names: conditional VaR (CVaR) in the optimization literature and in deep hedging, and tail VaR in insurance. [TN38] derives the representation that makes it optimizable. Example 38.1 computes the two measures side by side on a normal book.

Example 38.1 The two measures on the same book

Consider a \$100 million position with a 2.0% daily volatility (a daily standard deviation of \$2.0 million) and normally distributed returns. Under normality both measures are multiples of the standard deviation σ: $\mathrm{VaR}_c = z\sigma$ and $\mathrm{ES}_c = \sigma N'(z)/(1-c)$, where $z = N^{-1}(c)$ is the standard normal quantile at confidence c and N' is the standard normal density:

Confidence	z	VaR	ES	ES/VaR
95.0%	1.6449	\$3.29m	\$4.13m	1.254
97.5%	1.9600	\$3.92m	\$4.68m	1.193
99.0%	2.3263	\$4.65m	\$5.33m	1.146
99.9%	3.0902	\$6.18m	\$6.73m	1.090

The ratio falls as the confidence rises, because the normal tail thins so fast that conditioning on being past a distant point adds little. That ratio is a property of the normal distribution and not of the portfolio, which is exactly why it cannot be trusted: on a fatter tail it widens, and Example 38.6 shows by how much.

The regulatory switch reads directly off this table. The Basel framework of Chapter 39 replaced 99% VaR with 97.5% ES, and under normality the two nearly coincide:

$$\mathrm{VaR}_{99\%} = 2.3263\,\sigma, \qquad \mathrm{ES}_{97.5\%} = 2.3378\,\sigma, \qquad \text{a ratio of } 1.0049.$$

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The confidence level was chosen to make the change roughly capital-neutral for a normal book, so that what moved was not the level of the requirement but its sensitivity to the shape of the tail. Taken on its own, the switch from VaR to ES charges a bank with a genuinely fat-tailed book more and a bank with a normal book the same; the framework changed other things at the same time, which Chapter 39 sets out. Nor does it buy precision. Estimated from 500 days, a 97.5% ES averages only about a dozen widely dispersed losses, and on a fat-tailed book its standard error is about 14% of its value ([TN38]), no better than that of the 99% VaR it replaced, which Section 38.5 computes.

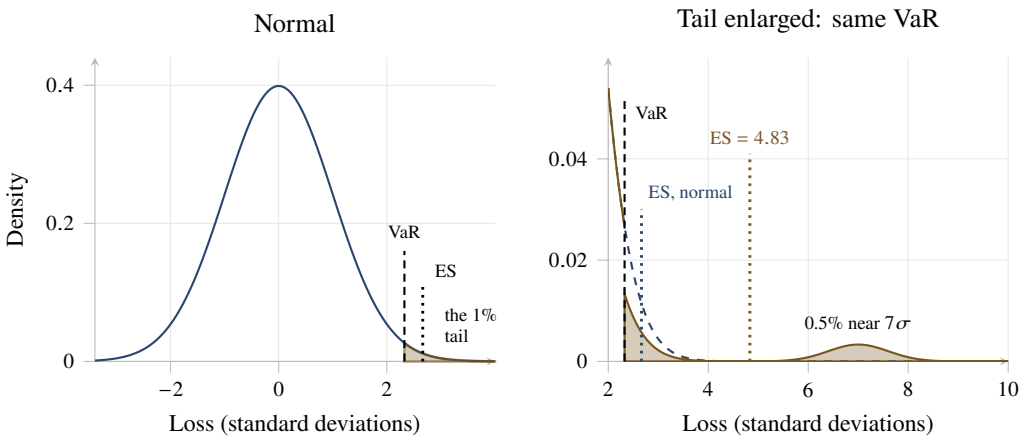


Figure 38.1: Losses at 99%, in standard deviations. Left: the standard normal. Right, the tail enlarged: the same distribution below VaR, with half its 1% tail moved to a cluster near 7σ. Illustrative.

Figure 38.1 carries the whole distinction. VaR is the dashed line: it locates the shaded region's left edge and is indifferent to everything inside it. Two portfolios with identical VaR can have any expected shortfall above that VaR, because the shading can be redistributed arbitrarily without moving the edge. The right panel does exactly that: the two distributions agree up to the VaR, so the VaR is the same 2.3263σ, but half of the 1% tail has been moved to a cluster of losses near 7σ (the profile of a book that has sold deep out-of-the-money protection), and the expected shortfall nearly doubles, from 2.67σ to 4.83σ, the average of the half tail left in place and the cluster at 7σ. A trader whose limit is set in VaR can exploit that, and Section 38.4 shows the exploitation is not even subtle.

38.4 COHERENCE, AND WHERE VAR FAILS

A risk measure ρ is called **coherent** if it has four properties, each a piece of common sense about risk:

- **Monotonicity.** A position that loses more than another in every outcome carries at least as much risk.
- **Positive homogeneity.** Doubling a position doubles its risk.
- **Translation invariance.** Adding an amount of cash to a position reduces its risk by exactly that amount.
- **Subadditivity.** Two positions A and B held together carry no more risk than the two carry separately:

$$\rho(A + B) \;\le\; \rho(A) + \rho(B). \tag{38.3}$$

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Subadditivity, the property that matters here, is the formal statement that diversification cannot hurt. It is also what makes a measure usable as a limit: if (38.3) fails, desk limits do not aggregate, and a firm can be within every limit and over its own total.

Expected shortfall satisfies (38.3) always; [TN38] proves this and the other three properties. VaR does not, and the counterexample needs no exotic structure at all: Example 38.2 builds it from two bonds.

Example 38.2 Two bonds that break the measure

A book holds two independent corporate bonds, \$10 million each, with a 0.8% default probability and zero recovery. Measure at 99%.

Each bond alone. A loss occurs with probability 0.008, which is *less* than the 1% tail being measured. The no-default branch carries 99.2% of the probability, so the quantile is zero up to that level and jumps to the full loss beyond it:

$$\mathrm{VaR}_c(\text{one bond}) = \begin{cases} \$0, & c \le 99.2\%, \\ \$10\text{m}, & c > 99.2\%. \end{cases}$$

At 99% the confidence level sits just below the jump, so $\mathrm{VaR}_{99\%}(\text{one bond}) = \$0$.

The two together. The loss distribution is:

Loss	Probability	Cumulative
\$0m	0.984064	0.984064
\$10m	0.015872	0.999936
\$20m	0.000064	1.000000

Since $0.984064 < 0.99$, the 99th percentile falls in the \$10m outcome:

$$\mathrm{VaR}_{99\%}(\text{both}) = \$10\text{m} \;>\; \$0 + \$0 = \mathrm{VaR}_{99\%}(A) + \mathrm{VaR}_{99\%}(B).$$

Diversifying made the measured risk go from nothing to \$10 million. Nothing pathological was constructed: two ordinary bonds, independent, with a default probability smaller than the tail being measured.

Expected shortfall behaves. It averages the worst 1% rather than reading its boundary. For one bond the worst 1% is the 0.8% default branch at \$10m and 0.2% of the no-loss branch, so $\mathrm{ES}_{99\%} = (0.008 \times 10 + 0.002 \times 0)/0.01 = \$8\text{m}$, against a VaR of zero. For the pair:

$$\mathrm{ES}_{99\%}(\text{both}) = \frac{(1 - 0.999936) \times 20 + (0.999936 - 0.99) \times 10}{0.01} = \$10.06\text{m} \;\le\; \$16\text{m}.$$

The mechanism is worth naming because it generalizes. VaR fails most readily where the loss distribution has a *jump* near the chosen quantile, which describes a credit portfolio of few issuers, every book with digital or barrier payoffs, and every position with a default or knock-out. It can fail on continuous distributions too, if they are skewed or fat-tailed enough, and it is guaranteed to be subadditive at any confidence level above one half when losses are **elliptical**: jointly distributed so that every portfolio's loss has the same shape, differing only in location and scale, as jointly normal and jointly Student's t losses do. So it is well behaved on smooth, roughly elliptical distributions and unreliable on precisely the books that need measuring.

VaR's blindness beyond the quantile also creates a perverse incentive. A desk measured on VaR can sell far-out-of-the-money options, collect premium, and report little or no additional risk, because the large loss has a probability below $1 - c$ and so lies beyond the quantile. Expected shortfall counts that loss at its probability-weighted size, which is a practical argument for the switch, and Section 38.8 gives a sharper one.

38.5 HISTORICAL SIMULATION

The dominant method is also the simplest. Take the last n trading days (500, about two years, is common, and the Basel rules require at least one year) and apply each day's market moves

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to *today's* portfolio, producing n hypothetical profits and losses. The measure is read off that sample.

What it assumes is one thing and it is a large one: that the last n days are a fair sample of tomorrow. It makes no distributional assumption at all, which is its great advantage: fat tails, skew, and the actual joint behavior of markets are captured without being modeled.

What it cannot do is produce a loss that has not happened. A 500-day window at 99% is estimating from five observations, and it cannot represent a move larger than the largest in the sample. A market that has been calm for two years reports a small number regardless of what is in the book, which is Section 38.18's problem and the reason the Basel capital framework does not rely on this window alone.

Weighting the observations helps but does not solve the problem. Exponentially weighting recent days makes the measure responsive, at the cost of shrinking the effective sample, which a tail estimate can least afford. Example 38.3 reads both measures off a 500-day window and then weights the same window twice, first with the bad days old, then with them recent.

Example 38.3 Five losses out of five hundred

The book is $10 million in an equity index, scored against the last 500 trading days. The 500 daily returns are simulated for illustration, not market data: Student's t with four degrees of freedom, scaled to a 1.25% daily standard deviation, the fat-tailed shape Section 38.8 examines. Each day's return applied to today's $10 million gives one scenario loss. The six worst losses are:

		As recorded		Calendar reversed	
Rank	**Loss ($000)**	**Days ago**	**Weight**	**Days ago**	**Weight**
1	581	428	0.064%	73	0.379%
2	437	458	0.055%	43	0.441%
3	424	446	0.059%	55	0.415%
4	408	249	0.157%	252	0.155%
5	396	442	0.060%	59	0.407%
6	357	383	0.080%	118	0.303%

With equal weights each scenario carries 0.2%, so the five worst losses are exactly the 1% tail. The 99% VaR is the loss with exactly five scenarios beyond it, the sixth worst, which is how (38.1) reads on a sample of 500: $\text{VaR}_{99\%} = \$357{,}000$. Expected shortfall is the mean of the five losses beyond it:

$$\text{ES}_{99\%} = \frac{581 + 437 + 424 + 408 + 396}{5} = \$449{,}000.$$

Some systems report the fifth-worst loss, $396,000, instead. That the convention alone moves the number by 11% is a measure of how little five observations are.

Now weight by recency, giving the scenario i days ago the weight

$$\pi_i = \frac{\lambda^i}{\sum_{j=1}^{500} \lambda^j}, \qquad \lambda = 0.995,$$

so that yesterday counts 0.544% and the oldest day 0.045%. The VaR is then the first loss, counting down from the worst, at which the running weight passes 1%. Four of the five worst losses are a cluster 428 to 458 days old, and the five together carry only 0.394%. The running weight does not pass 1% until the twelfth-worst loss, and the weighted VaR is $287,000, 20% below the equal-weighted figure, because the window's bad episode is old. Reverse the calendar, so that the same cluster is 43 to 73 days old, and the three worst losses alone carry 1.24%: the weighted VaR is the third-worst loss, $424,000, 19% above.

The weights moved the VaR between $287,000 and $424,000 without adding a scenario. They can only choose among losses already in the window, so whatever the weights, neither measure can

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exceed \$581,000, the largest loss in the sample. A window whose worst day was mild reports a mild number however its days are weighted.

The weighting in the example is the scheme of Boudoukh, Richardson and Whitelaw (1998). A second refinement rescales the scenarios to today's volatility instead of reweighting them, and Section 38.10 takes it up once the volatility estimators it needs are in place.

Five observations are also a small sample in a precise sense. An estimated quantile has a sampling standard error of approximately

$$\text{s.e.}\left(\widehat{\text{VaR}}_c\right) \approx \frac{1}{f(\text{VaR}_c)}\sqrt{\frac{c\,(1-c)}{n}}.$$

The square root is the standard error of a proportion: the fraction of the n days that land beyond the estimate is uncertain by that much. Dividing by $f(\text{VaR}_c)$, the density of the loss at the quantile, converts that uncertainty in probability into uncertainty in dollars, and the tail is where the density is lowest, so a small error in probability becomes a large error in the loss.

For the book of Example 38.3, the daily loss on the \$10 million position has the example's 1.25% standard deviation, \$125 thousand, and is a standard t_4 variable rescaled to it. A standard t_4 has variance 2, so the scale, in thousands of dollars, is $125 \times \sqrt{1/2} = 88.39$. The t_4's 99% quantile is 3.7469, and its density is $f_4(t) = \frac{3}{8}(1 + t^2/4)^{-5/2}$. The loss's own density is that density divided by 88.39: stretching a variable by a factor spreads the same probability over a range that much wider, so the probability per thousand dollars is that much smaller. With $c = 0.99$ and $n = 500$, the calculation runs

$$\begin{aligned}
\sqrt{c\,(1-c)/n} &= \sqrt{0.99 \times 0.01/500} = 0.004450,\\
\text{VaR}_{99\%} &= 88.39 \times 3.7469 = \$331.2 \text{ thousand},\\
f(\text{VaR}_{99\%}) &= \frac{f_4(3.7469)}{88.39} = \frac{0.008682}{88.39} = 0.0000982 \text{ per thousand dollars},\\
\text{s.e.} &= \frac{0.004450}{0.0000982} = \$45.3 \text{ thousand},
\end{aligned}$$

14% of the true VaR. The example's \$357,000 is a little over half a standard error from the truth, well within what a 500-day window can resolve.

In practice the density is not known. The **bootstrap** estimates the standard error without it: draw many new samples of 500 from the window itself, with replacement, recompute the VaR on each, and take the standard deviation of the results. For a quantile this far out the answer is rough, because every resample is built from the same few tail losses and none can contain a loss larger than the largest of them.

38.6 THE MODEL-BUILDING APPROACH

The alternative assumes a distribution and computes. For a portfolio linear in its risk factors, with position vector $\mathbf{w}$ and factor covariance Σ,

$$\text{VaR}_c = z\sqrt{\mathbf{w}^\top \Sigma\, \mathbf{w}}, \tag{38.4}$$

where z is the standard normal quantile at confidence c.

Equation (38.4) is fast, transparent, and differentiable, which is why it survives despite its assumptions: Section 38.15 needs that derivative and historical simulation does not supply one cleanly. Example 38.4 applies it to two stocks.

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Example 38.4 Two stocks

A book holds \$10 million of stock 1, with a 2.0% daily volatility, and \$5 million of stock 2, with 1.0%; the correlation between them is 0.3. In dollars the daily standard deviations are \$0.200m and \$0.050m, and the square root in (38.4) is the portfolio's daily standard deviation:

$$\sigma_P = \sqrt{0.200^2 + 0.050^2 + 2 \times 0.3 \times 0.200 \times 0.050} = \sqrt{0.0485} = \$0.2202\text{m}.$$

At 99%, $z = 2.3263$, so the one-day VaR is $2.3263 \times 0.2202 = \$0.512$m. Over ten days the square-root-of-time rule of Section 38.14 multiplies by $\sqrt{10}$, giving \$1.620m. It is a rule, not a result: it holds only if daily returns are independent, and that section shows what happens when they are not.

Standalone, the two positions have VaRs of $2.3263 \times 0.200 = \$0.465$m and $2.3263 \times 0.050 = \$0.116$m, which sum to \$0.582m. The portfolio's \$0.512m is \$0.069m less, a diversification benefit of 11.9% that exists only because the correlation is below one.

Equation (38.4) makes two assumptions, and both are about shape. It assumes normality, which Example 38.6 measures; and it assumes the book is linear in its factors, which fails for any book with options in it. The usual repair for the second is a **delta–gamma** expansion, which adds a quadratic term so that convexity is at least approximated. Example 38.5 applies it.

Example 38.5 Sold puts on the same stock

Keep the book of Example 38.4, but hold the position in stock 1 through sold put options instead, with the same dollar delta of \$10 million and a dollar gamma of −\$100 million. A sold put gains as the stock rises, so its delta is positive, and like any sold option it has negative gamma. (The dollar delta $\Delta_\$$ and dollar gamma $\Gamma_\$$ are the delta times the stock price and the gamma times its square.) A return r on the stock then changes the position's value by approximately

$$\Delta V \approx \Delta_\$ \, r + \frac{1}{2} \Gamma_\$ \, r^2 = 10\, r - 50\, r^2 \qquad (\$ \text{ millions}).$$

This is Chapter 18's expansion, equation (18.1), restricted to its stock terms and written per unit return: with $\Delta S = S\, r$, its delta term $\Delta\, \Delta S$ becomes $\Delta_\$ \, r$ and its gamma term becomes $\frac{1}{2}\Gamma_\$ \, r^2$. Taken alone, the stock's 99% fall is $r = -2.3263 \times 2.0\% = -4.653\%$. The linear term loses \$0.4653m, the standalone figure above; the quadratic term adds $50\, r^2 = \$0.1082$m, for a loss of \$0.5735m, 23% more than the linear term alone. The loss grows as the stock falls, and a rise produces a loss only beyond 20%, ten standard deviations away, so the loss at the 99% fall is the position's 99% VaR under the expansion. Equation (38.4) would still report \$0.465m, because it sees the dollar delta and nothing else.

Two terms suffice here because the sold puts' gamma, large as it is, changes little over a 4.65% move, so the parabola built from today's delta and gamma stays close to the position's true value. The approximation weakens when gamma itself changes quickly (an option near expiration with the stock close to its strike) and fails outright at a barrier. Had the puts carried a knock-out barrier 3% below the spot, the position would change form as the stock crossed it, and an expansion built from today's delta and gamma could not see a change it had not yet reached.

A portfolio of barriers and autocallables from Chapters 34 and 35 is therefore not well approximated by two terms, and for those books full revaluation under Chapter 20's Monte Carlo is the only reliable route. Section 38.7 sets out that route, and a shortcut for the smooth option books in between.

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38.7 NONLINEAR BOOKS: CORNISH–FISHER AND MONTE CARLO †

Between the linear book of Example 38.4 and a book of barriers lies most of what a derivatives desk holds: options whose value is curved but smooth. Two methods serve them, one analytical and one by simulation, and Table 38.1 sets them beside the two already met.

Where the quadratic approximation is good, its loss distribution can be summarized without simulation. With normally distributed factors, a delta–gamma loss L, with mean μ and standard deviation σ, has known higher moments too:

- its **skewness** $\gamma_1 = \mathbb{E}[(L-\mu)^3]/\sigma^3$ measures asymmetry: zero for a symmetric distribution, positive when the long tail lies on the loss side;
- its **excess kurtosis** $\gamma_2 = \mathbb{E}[(L-\mu)^4]/\sigma^4 - 3$ measures how much heavier its tails are than a normal distribution's, for which it is zero.

The **Cornish–Fisher expansion** turns these moments into a quantile by correcting the normal one:

$$z_{\mathrm{CF}} = z + \frac{z^2-1}{6}\,\gamma_1 + \frac{z^3-3z}{24}\,\gamma_2 - \frac{2z^3-5z}{36}\,\gamma_1^2, \qquad \mathrm{VaR} \approx \mu + z_{\mathrm{CF}}\,\sigma,$$

where $z = N^{-1}(c)$ is the standard normal quantile at the confidence level, 2.3263 at 99%. For the sold puts of Example 38.5, write the return as $r = 0.02Z$, Z standard normal; the loss $-10r + 50r^2$ is then $-aZ + bZ^2$ with $a = 10 \times 0.02 = 0.2$ and $b = 50 \times 0.02^2 = 0.02$ (\$m), and its moments follow from those of Z:

$$\mu = b = 0.020, \quad \sigma = \sqrt{a^2+2b^2} = 0.2020, \quad \gamma_1 = \frac{6a^2b + 8b^3}{\sigma^3} = 0.590, \quad \gamma_2 = \frac{48\,b^2(a^2+b^2)}{\sigma^4} = 0.466.$$

A normal quantile on μ and σ alone gives \$0.490m; the skewness term lifts it to \$0.578m and the fourth-order terms, kurtosis and squared skewness, bring it to \$0.573m, against the example's exact \$0.5735m ([TN38]). The expansion is reliable for moderate skewness like this and can misbehave for strongly nonlinear books, which is where simulation takes over.

Monte Carlo VaR draws the risk factors from their joint distribution many times, values the portfolio in each draw, and reads the quantile and the tail mean off the simulated losses. In **full revaluation** each draw reprices every position with its own pricing model; in **partial simulation** each draw applies the delta–gamma approximation, which is faster and inherits that approximation's limits.

Either way the result carries sampling error, and the standard-error formula of Section 38.5 measures it, with the number of draws M in place of the number of days. For the sold puts, the loss density at the VaR is the stock return's density there divided by the rate at which the loss grows with the return, which is $|-10 + 100\,r| = 14.653$ at $r = -4.653\%$, so the density is

$$f(\mathrm{VaR}_{99\%}) = \frac{N'(2.3263)}{0.02 \times 14.653} = 0.0909 \text{ per million dollars.}$$

With $M = 10{,}000$ draws the standard error is

$$\text{s.e.} = \frac{\sqrt{0.99 \times 0.01/10{,}000}}{0.0909} = \$0.011\text{m},$$

1.9% of the \$0.5735m VaR, and with 100,000 draws it is 0.6%. Unlike a historical window, a simulation can be lengthened as far as the machine allows, so its error is a cost of computing rather than a shortage of data.

The table sets out the trade-off. Below the first row, each step down handles curvature more faithfully and costs more computing; historical simulation, which already reprices in every scenario, alone takes its distribution from data rather than from an assumption, and Monte

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Method	Distribution	Options	Typical users
Historical simulation	the window's, as observed	repriced in each day's scenario	bank trading desks; clearing houses, volatility-scaled
Variance–covariance	normal	delta only	limit and allocation systems; fund VaR limits
Delta–gamma with Cornish–Fisher	normal factors	quadratic approximation	quick checks on smooth option books
Monte Carlo, full revaluation	any that can be simulated	repriced in each draw	exotic and structured books; insurers' internal models

Table 38.1: Four ways to compute VaR or expected shortfall.

Carlo alone can be made as precise as the budget allows. Section 38.20 says why each user ends up where the last column puts it.

38.8 FAT TAILS, AND A COUNTERINTUITIVE RESULT

The normal distribution is a poor description of daily returns: large moves occur far more often than it allows. The standard alternative is **Student's t distribution** with ν degrees of freedom, a bell curve whose density falls off as a power of the loss, $|x|^{-(\nu+1)}$, rather than exponentially in its square. The fewer the degrees of freedom, the fatter the tails, and as ν grows the t approaches the normal. Its variance is $\nu/(\nu-2)$ for $\nu > 2$, so a t variable multiplied by $\sqrt{(\nu-2)/\nu}$ has unit variance and can then be scaled to any chosen standard deviation; for $\nu = 4$, a common fit to daily equity returns, the factor is $\sqrt{1/2}$, which is where Section 38.5's 88.39 came from. Example 38.6 uses it to change a book's shape while holding its volatility fixed.

Example 38.6 The same volatility, a different shape

Take the \$100 million book of Example 38.1 and change nothing except the distribution: Student's t with four degrees of freedom, rescaled to the *identical* 2.0% daily standard deviation. The mean and volatility are the same; only the tail is fatter.

	VaR		ES		
Conf.	Normal	Fat-tailed	Normal	Fat-tailed	ES rise
95.0%	\$3.29m	\$3.01m	\$4.13m	\$4.53m	+10%
99.0%	\$4.65m	\$5.30m	\$5.33m	\$7.38m	+39%
99.9%	\$6.18m	\$10.14m	\$6.73m	\$13.70m	+103%

Figure 38.2 draws the two densities. The t is taller in the middle and thinner in the shoulders, and on the logarithmic scale of the right panel its tail lies orders of magnitude above the normal one; the vertical lines mark the two 99.9% VaRs, \$6.18m and \$10.14m.

Read the 95% row again. The fat-tailed VaR is *lower* than the normal one: \$3.01m against \$3.29m, some 8% *less* risk by that measure. Fat tails do not raise VaR at every confidence level. A distribution with the same variance and a heavier tail must be thinner somewhere, and at 95% that is where the quantile sits.

So a bank that fits a fat-tailed model and measures at 95% reports a risk reduction. As a quantile the number is right, but it ignores the tail, and the report does not say so.

Expected shortfall largely avoids this. It rises at every level in the table (10%, 39%, 103%) because it integrates the tail rather than sampling one point of it. It too falls below the normal at a low enough level, but its crossover sits near 89% against VaR's near 97.5%, so at the levels where risk is actually measured it moves the right way. This is a sharper argument for expected shortfall

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than subadditivity, because it needs no jump in the distribution: it happens on the most ordinary alternative distribution anyone would fit. It also shows what the normal assumption costs where it matters: at 99.9%, the level at which credit and default-risk capital is set, the normal model understates the true expected shortfall by a factor of two.

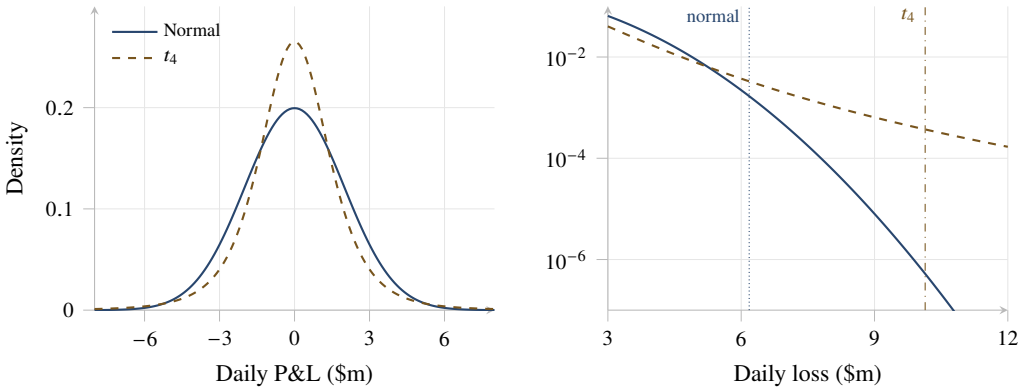


Figure 38.2: Daily P&L of the \$100m book under a normal and a t_4 with the same 2.0% standard deviation. Left: the densities. Right: the loss tail on a log scale, with each model's 99.9% VaR marked.

38.9 ESTIMATING A LOSS THAT HAS NOT HAPPENED †

Section 38.5 left a problem unanswered. Historical simulation cannot report a loss larger than the largest in its window, and a 99.9% measure of a daily loss asks about a percentile where a 500-day sample has half an observation.

Extreme value theory is the standard answer, and Example 38.7 measures what it achieves. The result that the excesses over a high threshold of almost any distribution follow a **generalized Pareto** law lets the tail be *fitted* from the observations that exist and then *extrapolated* past them.

Fix a high threshold u and look only at the losses beyond it. As u rises, the excess $Y = L - u$ of a loss over the threshold has, for almost any underlying distribution, approximately the distribution function

$$\mathbb{P}(Y \le y) = 1 - \left(1 + \frac{\xi\, y}{\psi}\right)^{-1/\xi},$$

with a scale ψ and a shape ξ that carries the tail's thickness:

- $\xi > 0$ is a power-law tail (a Student's t with ν degrees of freedom has $\xi = 1/\nu$, so the book of Example 38.6 has $\xi = 0.25$);
- $\xi = 0$ is an exponential-type tail, the limit for the normal, the lognormal and most other unbounded distributions whose tails die off faster than any power;
- $\xi < 0$ is a tail with an end point.

Fitting ψ and ξ to the losses above u and reading the quantile off the fitted law is the **peaks-over-threshold** method, and Example 38.7 sets it beside the two simpler estimates. [TN38] derives the results it rests on, from the threshold limit to the formulas the example uses.

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Example 38.7 Three ways to estimate a quantile you cannot observe

Take the fat-tailed book of Example 38.6: a daily loss in millions of dollars, distributed as a Student's t with four degrees of freedom and scaled to a standard deviation of 2.0, so that the true 99.9% VaR is \$10.14m and the expected shortfall \$13.70m. A risk manager sees only 500 days of losses drawn from it. One simulated window of 500 has mean loss $\bar{L} = -\$0.0011$m and standard deviation $s = \$1.7953$m, and its 26 largest losses, ranked from the largest down, are:

Ranks 1–5	6–10	11–15	16–20	21–26
7.2509	5.4179	3.7229	3.1511	3.0356
6.8962	4.6913	3.7217	3.1395	3.0329
6.1266	4.4446	3.5340	3.0841	3.0045
5.9353	4.1928	3.4639	3.0812	2.9119
5.7588	4.1022	3.3718	3.0789	2.8497
				2.8481

These losses, with $\bar{L}$ and s, are all the data the three estimates below use.

Normal fit. With $z = N^{-1}(0.999) = 3.0902$ from Example 38.1,

$$\text{VaR} = \bar{L} + z\,s = -0.0011 + 3.0902 \times 1.7953 = \$5.55\text{m}.$$

Empirical quantile. Placing the kth smallest of the 500 losses at percentile $(k-1)/499$ puts the 99.9th percentile at $k = 1 + 0.999 \times 499 = 499.501$, 50.1% of the way from the second-largest loss to the largest:

$$6.8962 + 0.501 \times (7.2509 - 6.8962) = \$7.07\text{m}.$$

Peaks over threshold. Set the threshold at the 26th-largest loss, $u = \$2.8481$m, so that 25 losses, the worst 5%, exceed it. The generalized Pareto law is fitted to their excesses y_i over u by **maximum likelihood**: choose the ψ and ξ under which the observed excesses are most probable, which means maximizing the logarithm of their joint density,

$$\ln \mathcal{L}(\psi, \xi) = -25 \ln \psi - \left(1 + \frac{1}{\xi}\right) \sum_{i=1}^{25} \ln\left(1 + \frac{\xi\, y_i}{\psi}\right),$$

over $\psi > 0$ and $\xi \geq -1/2$ (below $-1/2$ the maximum-likelihood estimates lose their usual large-sample behavior). That gives $\hat{\psi} = 1.1511$ and $\hat{\xi} = 0.0983$. Since 25 of the 500 days exceed u, a probability of $1 - c = 0.001$ is a fraction $(500/25) \times 0.001 = 0.02$ of the excesses, and solving the fitted law for that fraction gives

$$\text{VaR} = u + \frac{\psi}{\xi}\left[\left(\frac{n}{n_u}(1-c)\right)^{-\xi} - 1\right] = 2.8481 + \frac{1.1511}{0.0983}\left(0.02^{-0.0983} - 1\right) = \$8.34\text{m},$$

with $n = 500$ days, $n_u = 25$ exceedances and $c = 0.999$. Averaging the fitted tail beyond that point gives the expected shortfall:

$$\text{ES} = \frac{\text{VaR} + \psi - \xi u}{1 - \xi} = \frac{8.3389 + 1.1511 - 0.0983 \times 2.8481}{1 - 0.0983} = \$10.21\text{m}.$$

All three VaR estimates fall short of the true \$10.14m. The window's largest loss, \$7.25m, is below it, so the empirical quantile cannot reach it, and the 25 excesses suggest a thinner tail than the true one ($\hat{\xi} = 0.10$ against $\xi = 0.25$, or 0.19, the shape of the generalized Pareto law that best approximates the true t_4 excesses at this threshold, which [TN38] derives), so the extrapolation stops short too. Figure 38.3 draws the three against the data.

The figure shows what each method does where the data run out. The points stop at the largest loss, so the empirical quantile has nowhere further to go. The normal fit's tail falls away from the points almost at once and crosses the 0.001 line at \$5.55m. The generalized Pareto law follows the points and continues past them, reaching \$8.34m, but falls away faster than the

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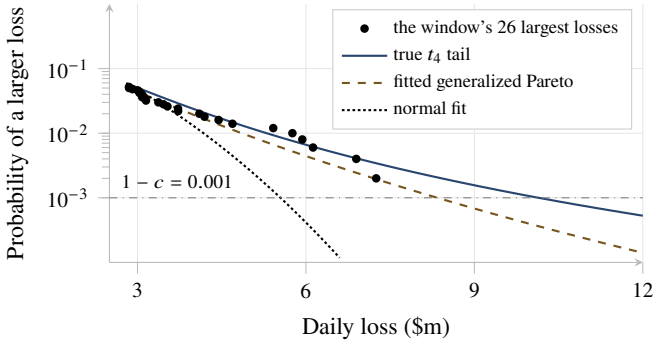


Figure 38.3: The tail of the window in Example 38.7, with the true distribution, the fitted generalized Pareto law and the normal fit. Each curve crosses the 0.001 line at its 99.9% VaR.

true tail, which does not cross until $10.14m.

One window shows how each calculation works, not how well each method performs. The table repeats the three calculations on 400 independent 500-day windows drawn from the same book, of which the example's is the first. The **mean VaR estimate** is the average of a method's 400 answers, the **bias** is that average divided by the true $10.14m, less one, and the **spread** is the standard deviation of the 400 answers.

Method	Mean VaR estimate	Bias	Spread across windows
Empirical quantile	$9.22m	−9.1%	$2.85m
Normal fit	$6.19m	**−39.0%**	**$0.70m**
Peaks over threshold	$10.11m	−0.3%	$3.32m

These are simulation results: the standard error of each mean is its spread divided by $\sqrt{400}$, under 2% of the true VaR for all three methods, and the 400 windows are drawn from a fixed seed. The normal fit is by far the most precise method and it is wrong by 39%. Precision and accuracy point in opposite directions here, and a risk report showing a stable number month after month is displaying the property that a normal assumption is good at producing. Peaks over threshold removes the bias, its −0.3% being well inside its standard error, and pays for it with the widest spread of the three. Its expected shortfall shows the cost: because the formula divides by $1 - \hat{\xi}$, the few windows whose fitted shape comes out high report very large values, so the median window's estimate, $11.48m, falls below the true $13.70m while the mean, $15.19m, is pulled above it.

The ceiling on historical simulation is not hypothetical. A window's largest loss stays below the true 99.9% VaR with probability $0.999^{500} = 61\%$, so in most windows no confidence level gets the empirical quantile to the right answer, because it lies off the end of the sample.

Two cautions keep this from being a solution. The threshold is a choice, and the estimate moves with it: a parameter the analyst picks, in a calculation the analyst is being judged by. And the method extrapolates the tail's *shape* from data in the tail's shoulder, which assumes the shape continues. It cannot see a mechanism that switches on only in the extreme, which is exactly what Section 38.11's correlations and Section 38.12's tail dependence describe.

So extreme value theory converts an impossible estimate into a difficult one. That is a real gain (a bias within sampling error of zero, against 39%), but it is not a known answer.

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38.10 ESTIMATING VOLATILITY

The model-building methods above need a volatility, and historical simulation can use one to rescale its scenarios. Two estimators dominate and they differ in one structural respect.

Exponentially weighted moving average (EWMA) weights past squared returns geometrically:

$$\sigma_n^2 = \lambda\sigma_{n-1}^2 + (1-\lambda)\, r_{n-1}^2, \tag{38.5}$$

where σ_n^2 is the variance estimate for day n, made at the end of day $n-1$, and r_{n-1} is the return on day $n-1$. The usual $\lambda = 0.94$ is the value J.P. Morgan's RiskMetrics (1996) chose because it minimized one-day forecast error across a broad set of daily series. The same recursion updates a covariance, with the product of two assets' returns in place of the squared return, which is how the correlations of Section 38.11 are usually kept current.

The **generalized autoregressive conditional heteroskedastic** (GARCH) model, in its GARCH(1,1) form, adds a constant, and the constant changes everything:

$$\sigma_n^2 = \omega + \alpha\, r_{n-1}^2 + \beta\, \sigma_{n-1}^2, \qquad V_L = \frac{\omega}{1-\alpha-\beta}, \tag{38.6}$$

where V_L is the long-run variance to which the forecast reverts. The three parameters are estimated by maximum likelihood on the return history, as the tail parameters were in Example 38.7: the chosen ω, α and β are those under which the observed returns are most probable given the variances the model assigns them. Setting $\omega = 0$ and $\alpha + \beta = 1$ in (38.6) recovers (38.5), so EWMA is GARCH with no long-run level to return to. That single difference decides the entire shape of the forecast, as Example 38.8 shows.

Example 38.8 One bad day, two forecasts

A market with a 1.00% daily volatility takes a −3% day.

Both estimators react almost identically. In percent squared, with $r_{n-1} = -3$ and GARCH's $\omega = 0.000002$ (0.02 in percent squared), $\alpha = 0.06$ and $\beta = 0.93$,

$$\sigma^2_{\text{EWMA}} = 0.94\times 1.00^2 + 0.06\times 3^2 = 1.48, \qquad \sigma^2_{\text{GARCH}} = 0.02 + 0.06\times 3^2 + 0.93\times 1.00^2 = 1.49,$$

for volatilities of 1.2166% and 1.2207%. A single day raises the estimate by about 22%.

They disagree completely about what happens next. GARCH has $\alpha + \beta = 0.99$ and therefore a long-run daily volatility of $\sqrt{0.000002/0.01} = 1.4142\%$, or 22.4% annualized. Writing σ_n^2 for the variance just estimated, the GARCH forecast of the variance k days later is

$$\mathbb{E}\big[\sigma_{n+k}^2\big] = V_L + (\alpha+\beta)^k\,\big(\sigma_n^2 - V_L\big),$$

so each day closes 1% of the remaining gap to the long-run level, and a deviation from it halves in about 69 trading days. Averaging that forecast over the T days to a horizon gives the variance

$$\bar{\sigma}_T^2 = V_L + \frac{1-(\alpha+\beta)^T}{T\,(1-\alpha-\beta)}\,\big(\sigma_n^2 - V_L\big),$$

which annualized is 19.4% for one day, 19.5% for ten days, 20.2% for three months and 21.4% for a year. EWMA's forecast is 19.3% at every horizon: its best guess for volatility in a year is today's estimate.

The two are fit for different jobs. A one-day margin or limit calculation cannot tell them apart, and EWMA's simplicity wins. Anything with a term structure needs the mean reversion: the volatility term structures of Chapter 19 slope the way they do because volatility reverts, and a model that denies reversion produces a flat term structure that the option market visibly contradicts.

Note also which way the current estimate sits. At 19.4% it is still below the model's 22.4% long-run level, and the GARCH forecast is *rising*. A risk system built on EWMA would report no such tendency, and the omission matters most just when volatility is about to rise.

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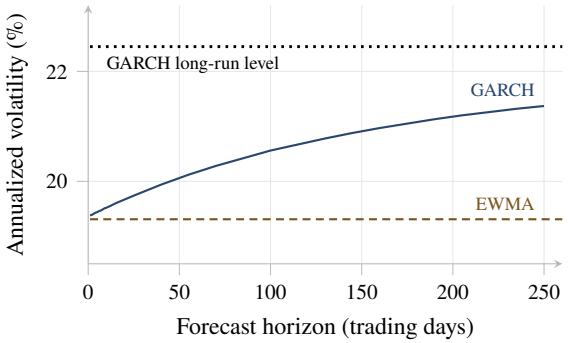


Figure 38.4: Average volatility to the horizon after the shock of Example 38.8.

Figure 38.4 is the argument in one picture. The gap between the two lines is not an estimation disagreement: both estimators saw the same data and agree about today. It is a disagreement about whether volatility has a level, and only one of them can produce a term structure.

Either estimator also repairs historical simulation, which Section 38.5 left reading every past day at the volatility it happened to have. **Volatility-scaled historical simulation**, proposed by Hull and White (1998), multiplies each day's return by $\sigma_{\text{today}}/\sigma_t$, today's volatility estimate over the estimate on the day it occurred. A calm-period day is scaled up when markets are volatile and a crisis day scaled down when they are calm, so the scenarios keep the historical shape at today's scale. Unlike weighting, rescaling can exceed the window's largest loss. **Filtered historical simulation** (Barone-Adesi, Giannopoulos and Vosper 1999) goes a step further: it divides each historical return by that day's GARCH volatility and draws the resulting standardized shocks at random, passing them through the recursion (38.6) to build scenarios over several days. Clearing houses set margin with volatility-scaled scenarios (Section 38.20).

38.11 CORRELATION, AND WHY IT IS THE WEAK INPUT

Example 38.8's estimators supply the diagonal of the covariance matrix in (38.4). The rest of it is harder. A volatility is pinned down by every day's return, large or small; a correlation depends on how the two series move together, and the ordinary days that dominate any window say little about how they move together on a stressed one. Correlations are therefore the least stable input in (38.4). Example 38.9 shows how far the measure moves with one.

Example 38.9 One number, revalued

Position A is \$60m at 1.0% daily volatility; position B is \$40m at 2.5%. In dollars the daily standard deviations are \$0.600m and \$1.000m, and at a correlation of 0.30

$$\sigma_P = \sqrt{0.600^2 + 1.000^2 + 2 \times 0.30 \times 0.600 \times 1.000} = \$1.311\text{m},$$

so the 99% one-day VaR is $2.3263 \times 1.311 = \$3.051$m. Change only the correlation:

Correlation	VaR	Change
−0.50	\$2.028m	−34%
0.00	\$2.713m	−11%
0.30	\$3.051m	—
0.60	\$3.355m	+10%
0.90	\$3.634m	+19%
1.00	\$3.722m	+22%

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A move from 0.30 to 0.90 raises the reported risk by 19% with no change in any position and no change in any volatility.

Correlations also move most in exactly the episodes the measure exists to describe. Assets that are unrelated in normal conditions become one asset in a liquidation, because the common factor in a crisis is often not economic: it is that the same leveraged holders are selling all of it. So the input is estimated in calm periods and applied in violent ones, and for a book that is long both assets, as this one is, the error has a consistent sign: measured risk is too low.

This is not a call for conservative correlations. It is a reason the framework does not let a bank choose them freely, and Chapter 39's constrained diversification rules exist because of the sensitivity this table shows.

38.12 COPULAS AND TAIL DEPENDENCE

Correlation summarizes dependence in one number, and one number is not enough. Two markets can have identical correlation and behave completely differently in the tail, which is what a copula (Section 24.13) separates out: the marginal distributions on one side, the dependence structure on the other. Example 38.10 holds the margins and the correlation fixed and changes only the copula.

Example 38.10 Same margins, same correlation, different joint extremes

A book holds two exposures, each normally distributed, with correlation 0.50. The only thing that differs is the copula: Gaussian against Student's t with four degrees of freedom.

Probability both exceed	Gaussian copula	t_4 copula	Ratio
95th percentile	0.01219	0.01694	1.39×
99th percentile	0.001294	0.002877	2.22×
99.5th percentile	0.0004963	0.001385	2.79×
99.9th percentile	0.00005426	0.0002635	**4.86**×

The discrepancy grows as the event gets worse, which is the opposite of what a modeler wants from an approximation. The limiting behavior is the cleanest statement. The coefficient of upper tail dependence (the limiting probability that one exposure is extreme given the other is) is exactly zero for a Gaussian copula at every correlation below one, and $2\,t_5\big(-\sqrt{5\times 0.5/1.5}\big) = 0.2532$ for this t copula, with t_5 the distribution function of Student's t with five degrees of freedom. Under a Gaussian copula, the chance that one exposure is in catastrophe given that the other is falls to zero far enough out, however correlated the assets are; under this t copula it never falls below a quarter.

Section 26.6 made this point about CDO tranches, where a Gaussian copula priced the senior tranches as if joint defaults could not cluster in the extreme. Here it decides something more basic: whether a portfolio risk measure gives simultaneous extreme losses their weight. A firm that models dependence with correlations alone has assumed the answer to the question it is asking.

Figure 38.5 shows where the difference lives. It draws 1,500 joint outcomes from each copula, and in the middle the two clouds are hard to tell apart. In the corners they are not: in this draw the t_4 copula puts 26 points beyond both 95th percentiles against the Gaussian's 16, and 7 beyond both 99th percentiles against 1. The two copulas agree in the body and diverge exactly where a risk measure is supposed to be doing its work.

38.13 FACTOR REDUCTION †

A swap book has sensitivities to twenty or more curve points, and (38.4) then needs a 20×20 covariance matrix estimated from data that barely supports it. The standard remedy is to replace the curve points with a few factors: Chapter 32's factor reduction, applied to a risk measure instead of a simulation.

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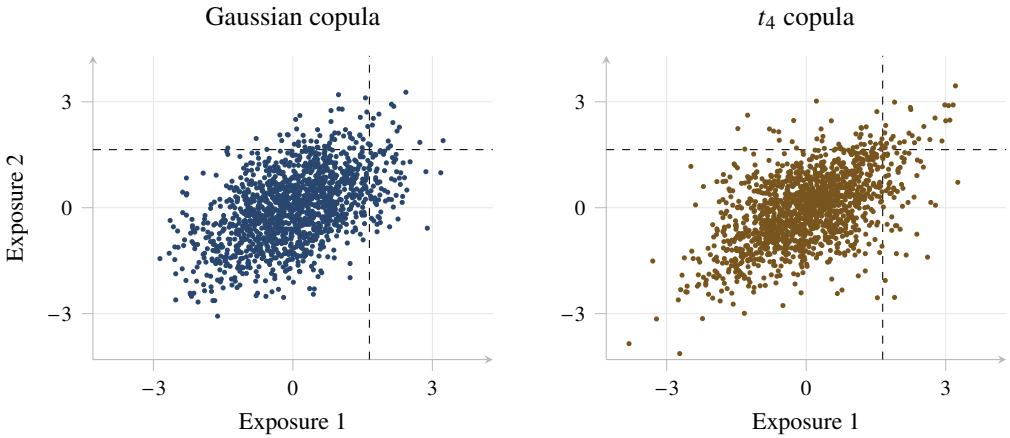


Figure 38.5: Joint outcomes, 1,500 under each copula, in standard deviations, with standard normal margins and correlation 0.50. Dashed lines mark each exposure's 95th percentile.

The factors are the principal components of the curve: the eigenvectors of Σ, the directions of movement that account, in order, for the largest shares of its variance. On a yield curve they are famously interpretable. Example 38.11 shows why interpretable is not the same as sufficient.

Example 38.11 Three factors, two portfolios, two verdicts

The curve has nine tenors from three months to thirty years. Their daily rate-change volatilities, and a directional book's sensitivities to each, are:

Tenor (years)	0.25	0.5	1	2	3	5	7	10	30
Volatility (bp a day)	3.5	4.4	5.3	6.0	6.2	6.0	5.7	5.4	4.8
Directional book ($000 per bp)	10	5	−8	15	12	−10	20	18	6

The correlation between the rate changes at tenors T_i and T_j is

$$\rho_{ij} = 0.7 + 0.3\, e^{-0.3\,|\ln T_i - \ln T_j|},$$

so neighboring tenors move almost together and the three-month and thirty-year rates have a correlation of 0.77. The model is illustrative rather than fitted; its short-end correlations are higher than daily Treasury data would give, which flatters the level factor. The covariance matrix is $\Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$, and its leading factors are:

Factor	**Variance explained**	**Cumulative**	**Shape**
Level	90.98%	90.98%	all tenors move together
Slope	3.94%	94.92%	short end against long end
Curvature	1.96%	96.88%	wings against the belly
Fourth	1.10%	97.97%	—

Three factors explain 96.88% of the variance of the curve, which is the number usually quoted and the reason three-factor models are standard.

Now measure two books against it. The first is the directional book above, with sensitivities of both signs scattered across the curve. For a vector $\mathbf{w}$ of sensitivities the full-covariance 99% one-day VaR is $2.3263\sqrt{\mathbf{w}^\top \Sigma\, \mathbf{w}}$, as in (38.4), which for this book is $800,607. The second is a **butterfly**, which on a yield curve means a position of one sign at two tenors, the wings, against the opposite sign at a tenor between them, the body. The name is borrowed from the option butterfly

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of Section 12.3, whose wings and body sit at strikes instead. Sized so that the sensitivities sum to zero, it is flat under a parallel shift and is a position on the curve's curvature, not its level. This one is long \$20,000 per basis point at two years and five years and short \$40,000 at three, with a full VaR of \$117,925. Keeping k factors replaces $\mathbf{w}^\top \Sigma\, \mathbf{w}$ by $\sum_{j \le k} \ell_j (\mathbf{v}_j^\top \mathbf{w})^2$, where ℓ_j and $\mathbf{v}_j$ are the eigenvalues and eigenvectors of Σ, and the table reports each book's VaR together with the share of its own variance that survives. (The VaRs use the unrounded quantile $N^{-1}(0.99) = 2.326348$; with 2.3263 the last digits move by up to \$16.)

	Directional book		Butterfly	
Factors kept	**VaR**	**Variance kept**	**VaR**	**Variance kept**
1	\$795,377	98.7%	\$20,390	3.0%
2	\$797,262	99.2%	\$23,225	3.9%
3	\$797,364	99.2%	\$38,194	10.5%
5	\$797,380	99.2%	\$62,434	28.0%
9	\$800,607	100%	\$117,925	100%

The same three factors that keep 99% of the first book's variance keep 10% of the second's. In VaR terms a three-factor system would report the butterfly at \$38,194 against an actual \$117,925, about a third of its risk, and it would do so while displaying the 96.88% figure as justification.

The two books sit on different parts of the decomposition, and Figure 38.6 shows it. The leading factors are smooth across the curve; the butterfly is three adjacent tenors with alternating signs, a shape none of them resembles, so it projects onto them weakly. The 96.88% is the share of the *curve's* variance in three factors. What a portfolio needs is the share of *its own* variance, and a book constructed to be neutral to level, and nearly neutral to slope as well, has deliberately projected itself onto the directions that were discarded. Factor adequacy is a property of the portfolio, not of the curve—and the portfolios most likely to fail the test are the ones relative-value desks hold.

That gives the practical rule. A risk system may reduce factors, but it must report the residual variance *for each book*, not for the market.

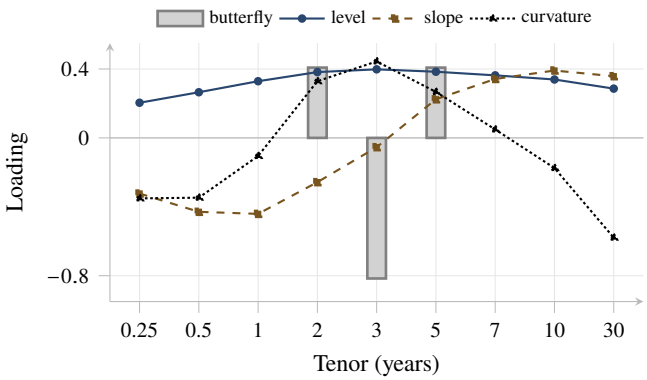


Figure 38.6: Loadings of the three leading factors of Example 38.11 by tenor, and the butterfly's sensitivities scaled to unit length.

38.14 SCALING ACROSS HORIZONS

Every figure so far has been a one-day measure, but capital and limits are often set over longer horizons, and the conversion is usually the square root of time:

$$\mathrm{VaR}_{h\text{-day}} \approx \sqrt{h}\, \mathrm{VaR}_{1\text{-day}}. \tag{38.7}$$

Equation (38.7) requires daily returns that are independent and identically distributed with

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mean zero, and for the quantile to scale like the standard deviation it also needs them normal; a fat-tailed one-day quantile does not scale by $\sqrt{h}$ even when the days are independent. The first condition is the one this section examines, because no one-day measure, and no one-day backtest, can reveal its failure. Suppose the correlation between returns k days apart is ρ^k, the pattern of a first-order autoregressive process with daily autocorrelation ρ. The variance of an h-day return is then the daily variance times $h + 2\sum_{k=1}^{h-1}(h-k)\rho^k$, one term for each pair of days, so under normality

$$\mathrm{VaR}_{h\text{-day}} = \mathrm{VaR}_{1\text{-day}}\sqrt{h + 2\sum_{k=1}^{h-1}(h-k)\,\rho^k}, \tag{38.8}$$

which reduces to (38.7) when $\rho = 0$. The table applies (38.8) to the one-day 99% VaR of \$4.65m from Example 38.1, scaled to ten days:

Daily autocorrelation	**10-day VaR**	**Error in (38.7)**
0.00	\$14.71m	—
0.10	\$16.10m	−9%
0.25	\$18.48m	−20%

The error column is the square-root figure's shortfall as a share of the true ten-day VaR, $(14.71 - \text{true})/\text{true}$: an autocorrelation of 0.10 leaves (38.7) about 9% too low and one of 0.25 a fifth too low, and positive autocorrelation is typical of the forced, one-directional selling of stressed markets, where the number is used. A mean-reverting market, with negative autocorrelation, breaks it the other way, so the rule is not conservative—it is merely wrong in a direction that depends on the market.

The deeper problem is that the horizon should not be a constant at all. Section 8.4's margin period of risk is set by how long a position takes to close, and that is a property of the position: a liquid futures hedge and a bespoke structured trade cannot share a horizon. Chapter 39's liquidity horizons exist to fix precisely this, replacing one number with several by risk-factor class.

38.15 WHERE THE RISK ACTUALLY SITS

A total is not a management tool. The useful question is which position is producing the risk, and the derivative of (38.4) answers it.

Marginal VaR is $\partial\mathrm{VaR}/\partial w_i$, the rate at which the VaR grows as position i grows. Differentiating (38.4),

$$\frac{\partial \mathrm{VaR}}{\partial w_i} = z\,\frac{(\Sigma\mathbf{w})_i}{\sigma_P}, \qquad \sigma_P = \sqrt{\mathbf{w}^\top \Sigma\,\mathbf{w}}, \tag{38.9}$$

where σ_P is the portfolio's standard deviation and $(\Sigma\mathbf{w})_i$ is the covariance of factor i's return with the portfolio's P&L. **Component VaR** is $w_i\,\partial\mathrm{VaR}/\partial w_i$, and because (38.4) is homogeneous of degree one in the positions, the components sum exactly to the total,

$$\sum_i \frac{w_i z(\Sigma\mathbf{w})_i}{\sigma_P} = z\sigma_P,$$

an Euler allocation (Euler's theorem for functions homogeneous of degree one), not an approximation. Example 38.12 applies it to the book of Example 38.9.

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Example 38.12 Sixty percent of the money, thirty percent of the risk

	Position	Standalone VaR	Component VaR	Share of risk
A	\$60m (60%)	\$1.396m	\$0.958m	31.4%
B	\$40m (40%)	\$2.326m	\$2.093m	**68.6%**
Total	\$100m	\$3.722m	\$3.051m	100%

B is 40% of the money and 69% of the risk, and the standalone numbers do not show this: they sum to \$3.722m against a portfolio VaR of \$3.051m, an 18.0% diversification benefit that has to be allocated somehow. Component VaR allocates it in the only way that both adds up and agrees with the marginal figures.

The marginal numbers are what a trader should be shown. In return units $\Sigma\mathbf{w} = (0.0090, 0.0295)$ and $\sigma_P = \$1.311\text{m}$, so (38.9) gives $2.3263 \times 0.0090/1.311 = 0.0160$ for A and $2.3263 \times 0.0295/1.311 = 0.0523$ for B. At the margin, a dollar moved into B adds 0.0523 of VaR against A's 0.0160, so shifting \$10m from A to B should add about $10 \times (0.0523 - 0.0160) = \$0.36\text{m}$, or 11.9%; revalued exactly, at \$50m each, the VaR is $2.3263\sqrt{0.500^2 + 1.250^2 + 2 \times 0.30 \times 0.500 \times 1.250} = \$3.441\text{m}$, 12.8% higher, while the book's size is unchanged. A limit on total VaR permits that trade as long as there is room under it; component VaR shows the whole rise came from B.

One caution applies, the chapter's recurring one: an exact allocation says nothing about whether the total is right. Component VaR sums exactly only because VaR is homogeneous; it does not require VaR to be subadditive, and Example 38.2 shows it is not. So the allocation can be exact and the total still wrong. Expected shortfall admits the same Euler decomposition and is coherent; under simulation its components are simply each position's average loss on the scenarios in the tail, $\mathbb{E}[L_i \mid L > \text{VaR}_c]$, which is one reason it is increasingly used for allocation as well.

Incremental VaR answers a third question: how much the VaR changes if a position is removed altogether. In the example, removing B leaves A's standalone \$1.396m, so B's incremental VaR is \$3.051m − \$1.396m = \$1.655m; removing A leaves \$2.326m, so A's is \$0.725m. The two sum to \$2.380m, not to the \$3.051m total, because each credits the whole \$0.671m diversification benefit to the position being removed, so the two together deduct it twice, once more than the total does: $\$3.722\text{m} - 2 \times \$0.671\text{m} = \$2.380\text{m}$. Incremental figures answer whether to keep or close a whole position; component figures are the ones that add up, and marginal figures the ones that guide a small trade.

38.16 BACKTESTING

A risk number is a testable prediction: at 99%, losses should exceed the VaR on 1% of days, and each day they do is an **exception**. Counting exceptions is the most direct validation any of this receives, and Example 38.13 shows how little a year of it establishes.

Example 38.13 Two hundred and fifty days is not much evidence

At 99% over 250 days, the expected number of exceptions is 2.5. With x exceptions in n days, a target exception rate $1 - c = 1\%$ and the observed rate $\hat{p} = x/n$, the Kupiec likelihood-ratio (LR) statistic is

$$\text{LR} = -2 \ln \frac{c^{n-x}\,(1-c)^{x}}{(1-\hat{p})^{n-x}\,\hat{p}^{x}},$$

which under a correct model is approximately chi-squared with one degree of freedom. The test rejects when it exceeds 3.841, the 95th percentile of that distribution, for a nominal 5% significance level; when, as here, few exceptions are expected, the exact binomial probabilities in the table are the alternative:

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Exceptions	Probability if correct	Kupiec LR	Reject?	Basel zone
0	0.0811	5.03	**yes**	green
1	0.2047	1.18	no	green
2	0.2574	0.11	no	green
3	0.2149	0.09	no	green
4	0.1341	0.77	no	green
5	0.0666	1.96	no	yellow
7	0.0097	5.50	**yes**	yellow
10	0.0002	12.96	**yes**	red

Anything from one to six exceptions is statistically indistinguishable from a correct model. Those counts are exception rates from 0.4% to 2.4%, so a model whose year looks like 97.6% coverage and one whose year looks like 99.6% both pass as 99% models.

Zero exceptions is a rejection too, and that is the row people forget: a model that never breaches is not conservative but miscalibrated, in the direction that wastes capital. The test is two-sided because a risk measure is a forecast, not a promise. The evidence is thin, though. A correct model shows no exceptions in 8.1% of years, and the 3.841 cutoff, a large-sample approximation that is coarse when only 2.5 exceptions are expected, rejects a correct model 9.5% of the time (zero exceptions, or seven or more) rather than the nominal 5%.

Note where the Basel zones, set in 1996, and the statistics disagree. Five exceptions moves a bank into the yellow zone, where the 1996 rules raised the multiplier applied to its VaR from 3 to between 3.40 and 3.85 (4 in the red zone); the current framework keeps the zones and applies them to a multiplier of 1.5 to 2.0, set out in Chapter 39. Yet at five exceptions the Kupiec statistic is 1.96 and does not reject, and at seven to nine the test rejects while the zone is still yellow. The zones are one-sided: they look only for too many exceptions, and they flag early. A correct model lands in yellow or worse in 10.8% of years, a rate the supervisor accepted in exchange for power: a model whose true coverage is only 97% is flagged in 87% of years, against 63% for a Kupiec rejection.

Figure 38.7 puts the two verdicts on one axis. The shaded band is the range the Kupiec test tolerates and the dashed rules are the Basel boundaries, and they do not line up: a bank with five exceptions has left the green zone and pays a higher multiplier while the statistic sees nothing worth reporting. Note also how much probability mass sits under a correct model at counts a supervisor would treat as a warning.

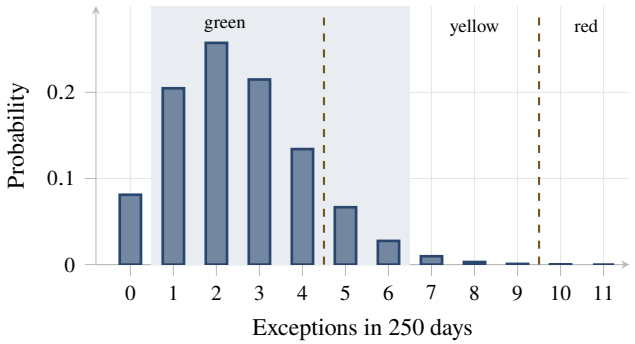


Figure 38.7: Distribution of 99% VaR exceptions in 250 days, Binomial(250, 0.01). Shading marks the range the Kupiec test does not reject; the dashed rules are the Basel zone boundaries.

Counting is only half of the validation. Kupiec asks whether the number of exceptions is right and cannot ask when they arrived, so a model that breaches on five consecutive days in one stressed week passes it exactly as a model that breaches five times spread through the

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year. Those two models are not equally good, and the second one is the one the bank wants: clustered exceptions mean the model's variance is stale, which is precisely the failure that matters, because it arrives when the losses do.

Christoffersen's conditional-coverage test adds the missing half. Alongside the coverage statistic it tests **independence** (whether an exception yesterday raises the chance of one today) by comparing the observed transition frequencies against the hypothesis that they do not, and the two statistics add, each with one degree of freedom, so the combined statistic is compared with a chi-squared distribution with two degrees of freedom (5.991 at 5%). A model can pass on coverage and fail on independence.

A clean exception count is weaker evidence than it looks. Chapter 39's supervisory framework counts exceptions and does not test their independence, so a desk that wants to know whether its model works has to run that test itself.

Expected shortfall cannot be backtested this way at all. There is no event to count: ES is a conditional mean, so validating it requires estimating a mean from the handful of observations in the tail. Tests built for it exist: Acerbi and Székely (2014) compare each tail loss with the forecast ES on the days VaR was exceeded. A separate obstacle concerns ranking competing forecasts rather than testing one: ES alone is not **elicitable**, meaning that no scoring rule's average is minimized by the true ES, so two ES forecasts cannot be ranked on their record alone; the pair of VaR and ES is elicitable, and [TN38] sets out both. The measure the framework moved to is the one that is harder to check, which is a real cost of the switch and the reason Chapter 39 backtests the VaR while capitalizing the ES.

38.17 STRESS TESTING

Every measure so far answers a question about probability: how large a loss is reached once in a hundred days, and how large the losses are beyond it. A **stress test** answers a different question. It revalues the portfolio under a specified scenario (a set of moves in prices, rates, volatilities, correlations and liquidity) and reports the loss, with no probability attached.

That is its weakness and its use. A number with no likelihood cannot be compared across books the way a VaR can, and it cannot be added to anything. But it does not depend on the window's estimated distribution, so it can describe events the window never contained and joint moves the estimated correlation matrix all but rules out. It targets the gaps Section 38.19 lists.

Scenarios come from two sources. **Historical scenarios** replay the moves of an actual episode on today's positions (October 19, 1987, the autumn of 2008, March 2020, the September 2022 gilt crisis of Section 8.9), and their merit is that nobody can call them implausible. **Hypothetical scenarios** are written by the firm for the risks it actually holds, and a good one moves correlations and liquidity as well as prices, because Section 38.11 showed that the joint behavior is what changes in a crisis. **Sensitivity analysis**, moving one risk factor at a time, is easier to run and to explain, and it misses exactly those joint moves. Example 38.14 sets a hypothetical scenario beside a sensitivity report on the same book.

Example 38.14 Two assets falling together

Take the book of Example 38.9: \$60m in asset A at a 1.0% daily volatility and \$40m in asset B at 2.5%, correlation 0.30, with a 99% one-day VaR of \$3.051m. The scenario is a liquidation day on which both assets fall by four of their own daily standard deviations, A by 4% and B by 10%.

The loss is

$$60 \times 4\% + 40 \times 10\% = \$2.4\text{m} + \$4.0\text{m} = \$6.4\text{m},$$

2.10 times the VaR. A sensitivity report that moves each asset alone by the same four standard deviations shows \$2.4m and \$4.0m, and its worst case, \$4.0m, falls 37.5% short.

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Under the model's own assumptions (normal returns, correlation 0.30), the portfolio's daily standard deviation is \$1.311m, so a \$6.4m loss lies 4.88 portfolio standard deviations out. Its probability is about 5 in 10 million a day, once in some 7,500 years of trading days. The stress test does not rely on that figure. It reports the loss and leaves its likelihood to judgment, because the figure is computed from the very correlation the scenario assumes has broken.

Reverse stress testing inverts the question. Instead of asking what a scenario costs, it fixes a loss (the one that would breach a limit, exhaust capital, or end the business) and asks which scenario produces it. It is the exercise most likely to find what the model misses, because it starts from the outcome rather than from the model's risk factors.

For a linear book with normally distributed factors the question has a closed form. Fix the loss at l. Many moves $\mathbf{r}$ of the factors lose exactly l on positions $\mathbf{w}$, and no single one has a probability, but they differ in how plausible they are: the joint density is highest close to the center of the distribution. Distance from the center is measured in standard deviations along whatever direction the move takes, by the **Mahalanobis distance** $\sqrt{\mathbf{r}^\top \Sigma^{-1} \mathbf{r}}$, and the most plausible move is the one that minimizes it subject to $\mathbf{w}^\top \mathbf{r} = -l$. The answer is

$$\mathbf{r}^* = -\, l \, \frac{\Sigma \mathbf{w}}{\mathbf{w}^\top \Sigma \, \mathbf{w}}, \tag{38.10}$$

and it lies l/σ_P standard deviations from the center, where $\sigma_P = \sqrt{\mathbf{w}^\top \Sigma \, \mathbf{w}}$ is the portfolio's standard deviation. Each factor moves in proportion to $(\Sigma \mathbf{w})_i$, its covariance with the portfolio, which is the same vector that gives marginal VaR in (38.9): the scenario loads most heavily on the positions with the highest marginal risk. For a nonlinear book there is no formula, and the scenario has to be found by searching over moves with full revaluation. Example 38.15 applies (38.10) to the book of Example 38.14.

Example 38.15 The most plausible way to lose \$6.4 million

Keep the book of Example 38.14 and fix the loss at the scenario's \$6.4m. In return units, $\Sigma \mathbf{w} = (0.0090,\ 0.0295)$ and $\mathbf{w}^\top \Sigma \, \mathbf{w} = 1.72$, so (38.10) gives

$$r_A^* = -6.4 \times \frac{0.0090}{1.72} = -3.35\%, \qquad r_B^* = -6.4 \times \frac{0.0295}{1.72} = -10.98\%.$$

The loss is $60 \times 3.35\% + 40 \times 10.98\% = \$6.4$m, as required, and the move lies $6.4/1.311 = 4.88$ standard deviations out.

The hand-written scenario of Example 38.14, four standard deviations down on each asset, lies $\sqrt{(4^2 + 4^2 - 2 \times 0.30 \times 4 \times 4)/(1 - 0.30^2)} = 4.96$ standard deviations out by the Mahalanobis distance. The joint density falls as $e^{-d^2/2}$ with that distance d, so the reverse stress test has found a route to the same loss at which the density is $e^{(4.961^2 - 4.880^2)/2} = 1.49$ times as high. It also says where the risk is: on the most plausible path B falls 4.39 of its standard deviations and A only 3.35 of its own, as B's 69% share of the VaR in Example 38.12 suggests. The scenario a risk manager would write by symmetry, both falling equally hard, is not the one that matters most.

Figure 38.8 draws the geometry. The ellipse is the set of moves as plausible as the reverse-stress scenario, the dashed line every move that loses exactly \$6.4m, and the scenario is where the two touch; any other point on the line, including the hand-written one, lies outside the ellipse, where the density is lower.

Stress tests and probability-based measures are used side by side rather than combined. A typical limit framework sets a VaR or ES limit and, separately, a ceiling on the loss under each of a handful of named scenarios; a breach of either requires action. Some firms attach rough probabilities to their scenarios to fold them into a single number, but that reintroduces

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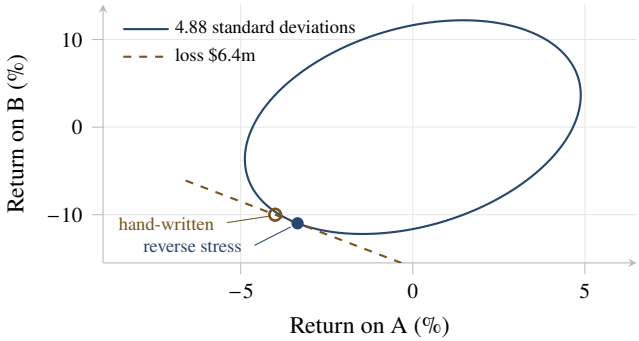


Figure 38.8: The most plausible move that loses $6.4m on the book of Example 38.14: the point where the iso-loss line touches a contour of the return distribution.

the estimate the stress test was built to avoid. The next section turns to the regulatory use of a crisis window, which keeps the probability machinery and changes only its data.

38.18 STRESSED MEASURES AND PROCYCLICALITY

Every estimator in this chapter is fitted to a window, and a window shrinks the measure as markets calm. Take a $100m position at 99%, one day: at a calm 1.0% daily volatility the VaR is $2.33m; estimated on a stressed window at 3.2%, it is $7.44m. The same book comes out a factor of 3.2 apart, and the difference lies entirely in the sample.

The consequence is perverse and it is structural. Risk-based capital falls through a calm period, permitting more risk to be taken, and rises after volatility arrives, forcing positions to be cut into a falling market. The measure amplifies the cycle it is meant to guard against—which the reader has already met as Section 8.9's collateral loop, appearing here as a capital spiral.

The repair is to add a measure that does not depend on the current window. A **stressed VaR**, added by Basel 2.5 in 2009, runs the same portfolio through a fixed twelve-month period of significant stress, so a bank's requirement cannot fall simply because the recent past was quiet. It is a crude device and it works, and Chapter 39 gives the current form.

What it does not fix is that any window, stressed or not, is history. A stressed window contains the crisis that happened, so it protects against the last crisis rather than the next one. And books are optimized against the measure, which means the positions that look cheap under it are precisely the ones whose risk it does not see.

38.19 WHAT THESE MEASURES DO NOT SEE

A risk measure describes a distribution, and not everything that matters about a book is in it.

Liquidity is absent. The measure assumes the position can be exited at the marked price; the gilt funds of Section 8.9, selling into a market their own sales were moving, could not.

Concentration is invisible. Two books with the same distribution have the same measure, whether the position is 1% of the market or 40% of it—and Amaranth, Archegos and the nickel short in Chapter 40 are all of the second kind.

Funding is a separate risk. A position can be correct, solvent, and still fail because variation margin must be paid before the offsetting gain is realized. Example 37.2's basis trade loses nothing and can still be closed for want of cash, and Chapter 40's Metallgesellschaft and LDI cases are the same mechanism at scale.

And the model itself is a risk. Every number here is conditional on the mapping from market factors to portfolio value being right, the mapping that Chapter 21's calibration and,

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increasingly, trained networks produce. Chapter 39 treats that as its own subject, because it is the one exposure the risk system cannot measure with a risk system.

Case Study 38.1 Twenty-five sigma, several days running

In August 2007 a number of quantitative equity funds lost heavily over the same few days. The chief financial officer of Goldman Sachs, asked to explain, said the firm's funds had been seeing twenty-five standard deviation moves, several days in a row.

Take the statement literally and it is not a description of an event. Under a normal distribution a twenty-five sigma move has a probability of roughly 10^{-138}. There have been about 10^4 trading days in a working lifetime and something like 10^{17} seconds since the universe began. Several such days in one week is not a rare event; it is a refutation of the distribution used to measure it.

So the sentence has two readings and only one survives. Either something astronomically improbable happened repeatedly, or the model assigning the probabilities was wrong. Section 38.8 has already measured how wrong a normal assumption can be in the tail, and this is that error stated by someone who had just paid for it.

What the measure could not see was crowding. The positions were not unusual individually. They were held, in similar amounts and the same direction, by many funds running similar signals, so the deleveraging of one marked the book of the next. The exposure was to other people's positions, and a measure calibrated on a firm's own history contains no variable for that, which is why concentration belongs in this section rather than among the estimators.

Backtesting could not have raised the alarm in time. Example 38.13 showed that a year of exceptions cannot reject a 99% model whose realized coverage looks like 97.6%, and that even a model whose true coverage is 97% stays in the green zone one year in eight. A book can run on a badly shaped distribution for a long while and produce an exception count no supervisor would question.

The practical rule is the one the episode states out loud. Backtesting checks the count; the test that matters is not whether the count of breaches looks reasonable but whether the shape is right. A model that reports twenty-five sigma has already told you it is the wrong model; no further evidence is required, and none should be waited for.

38.20 WHO USES THESE NUMBERS, AND WHY

Regulators use them to set capital, the use that decides the definitions: Chapter 39's rules are the reason the industry computes expected shortfall at 97.5% rather than VaR's 99%.

Boards and senior management use VaR because it is one number with a probability attached, and that is genuinely the reason it won: the measure that can be explained in a sentence displaced measures that could not, and its statistical defects were formalized only later.

Risk managers use component and marginal figures rather than totals, because the total is not actionable and the breakdown in Example 38.12 is.

Clearing houses and prime brokers use short-horizon measures to set margin, where the horizon is a close-out estimate and the confidence level is a business decision about how often to be wrong. The large swap clearing houses compute initial margin by volatility-scaled historical simulation, which the industry often calls filtered historical simulation, taking a VaR or an expected shortfall over a lookback of several years that includes a period of stress, so that a quiet year does not shrink the margin. LCH's SwapClear, the largest of them, uses expected shortfall at 99.7% on volatility-scaled scenarios from a ten-year lookback period, with a five-day close-out for house accounts.

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Insurers use the same idea at a different confidence and horizon. Solvency II, the European Union's insurance regime, sets the solvency capital requirement at the 99.5% VaR of an insurer's basic own funds, its assets less its liabilities, over one year, and an insurer with an approved internal model computes it by simulation.

Asset managers meet VaR as a limit on their funds. A European fund authorized under the rules for Undertakings for Collective Investment in Transferable Securities (UCITS) that measures its exposure by VaR must keep its 99%, twenty-day VaR below 20% of net asset value, or below twice that of a reference portfolio. In the United States, the SEC's Rule 18f-4, with which funds have had to comply since August 2022, imposes the same pair of limits on an open-end fund whose derivatives exposure exceeds 10% of its net assets.

Nonfinancial companies seldom manage by VaR, though some large ones disclose it, because their exposure is to operating cash flow rather than to the market value of a book. The variant they use is **cash-flow-at-risk**: the shortfall of a quarter's or a year's operating cash flow below its forecast that is exceeded with a stated probability.

Traders use them mainly as constraints, and a constraint is a thing to optimize against—which is why a measure's blind spots become positions. The London Whale of Section 40.10 is the plainest case: the measure changed and the book grew into it.

38.21 WHERE THIS CAME FROM

Value at risk was a practitioner construction that regulation adopted, and the theory arrived afterward to tell it what was wrong. Artzner, Delbaen, Eber and Heath (1999) sets out the axioms a risk measure should satisfy, shows that VaR violates subadditivity, and thereby names a defect desks had already met without being able to explain. Expected shortfall in its integral form satisfies them, as Acerbi and Tasche (2002) and Rockafellar and Uryasev (2002) showed for distributions with atoms, and the Basel framework moved to it in 2016, seventeen years after Artzner and his coauthors, an unusually clean case of a theoretical result changing a rule.

Embrechts, Klüppelberg and Mikosch (1997) is the extreme-value treatment behind Section 38.9's tail estimation, and McNeil, Frey and Embrechts (2015) is the modern synthesis.

SUMMARY

A risk measure turns a distribution into a number, and the choice of number is the whole subject. VaR is the quantile of (38.1): it says where the tail starts and nothing about what is in it, while expected shortfall averages the tail beyond that point. The Basel move from 99% VaR to 97.5% ES was calibrated to be capital-neutral under normality, so what changed was not the level but the sensitivity to tail shape.

That sensitivity is the case for ES. VaR can fail subadditivity where losses jump near the quantile: two ordinary bonds, each with a VaR of zero, can have a large VaR together. ES is coherent. The sharper argument is shape: a fat-tailed fit with the same volatility can lower VaR while raising ES, so the normal assumption understates exactly the tail ES measures.

Judgment enters through the inputs. EWMA is GARCH without a long-run level, so only GARCH produces a volatility term structure. Correlation is the weak input, and it moves most in the episodes the measure exists to describe; and correlation is not the whole of dependence, since a t copula makes joint extremes far more likely than a Gaussian one with the same correlation. Factor adequacy is a property of the portfolio, not of the curve, and allocation needs three figures, component, marginal and incremental VaR, because it asks three questions.

Estimation is where most of the error lives. Historical simulation is only as good as its window, and no reweighting adds information about a tail the window never saw; extreme value theory reaches beyond the data at a price in variance, and option books need a nonlinear

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treatment. Stress testing prices a named scenario with no probability attached, and reverse stress testing runs backward from a loss to the most plausible scenario that produces it.

Validation is weaker than anyone would like. At 99% over 250 days a wide range of exception counts is consistent with a correct model, and expected shortfall cannot be backtested by counting at all, which is why Chapter 39 backtests the VaR and capitalizes the ES. Because every estimator is fitted to a window, the requirement is smallest just before it is needed, which stressed measures exist to correct. Liquidity, concentration, funding and the model's own correctness lie outside any risk measure; Chapter 39 prices some and Chapter 40 shows the rest.

FURTHER READING

The coherence axioms and the subadditivity failure of Example 38.2 come from Artzner, Delbaen, Eber and Heath (1999), which is short and states the counterexample plainly; Acerbi and Tasche (2002) settles why the integral form of expected shortfall, and not the conditional mean, is the coherent one. On estimation, J.P. Morgan's RiskMetrics technical document (1996), which made EWMA the industry's default, states why $\lambda = 0.94$ and what it costs, and the GARCH literature, which begins with Bollerslev (1986), is best approached through its forecasting results rather than its estimation theory; Example 38.8's term structure is the part that matters for derivatives. On copulas, read Section 26.6's sources first: the tail-dependence result of Example 38.10 was known well before it was needed and ignored anyway, so read them for the history rather than the mathematics. On backtesting, Kupiec (1995) and the Basel Committee's 1996 backtesting framework are worth reading side by side precisely because they disagree, and the supervisory document sets out how its zones trade false alarms against the power to catch a bad model. Christoffersen (1998) is the paper for the other half of the question: it is short, and it is where the independence test of Section 38.16 comes from, together with the observation that coverage and independence are separable. Hull and White (1998) is the short paper behind volatility scaling, and Barone-Adesi, Giannopoulos and Vosper (1999) the one behind filtered historical simulation. On backtesting expected shortfall, Acerbi and Székely (2014) is the place to start, and Gneiting (2011) and Fissler and Ziegel (2016) supply the theory of why ES cannot be scored alone and how the pair with VaR can be. Chapter 39 gives the sources for the current capital framework, and [TN38] derives the representation that makes expected shortfall optimizable.

PRACTICE QUESTIONS

38.1 Using Example 38.1, state the 99% one-day VaR and ES for a \$100m book at 2% daily volatility, and explain why the ratio between them falls as confidence rises.

38.2 Show that 99% VaR and 97.5% ES nearly coincide for a normal loss, and explain what the regulator was trying to achieve by choosing those two levels.

38.3 State the subadditivity condition and explain why a measure that violates it cannot be used for desk limits.

38.4 Using Example 38.2, compute the 99% VaR of each bond alone and of the two together, and explain what feature of the loss distribution produces the failure.

38.5 Compute the 99% expected shortfall for the same two bonds, and verify that it satisfies subadditivity.

38.6 Explain why a desk measured on VaR has an incentive to sell far-out-of-the-money options, and what expected shortfall does to that incentive.

38.7 Explain what historical simulation assumes, and state the largest loss it is capable of reporting.

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38.8 Using Example 38.6, explain why the fat-tailed VaR at 95% is *lower* than the normal VaR, and why the expected shortfall is not.

38.9 Compute, from Example 38.6, the ratio of the fat-tailed to the normal expected shortfall at 99.9% and the same ratio for VaR, and explain why the expected shortfall error matters more than the VaR error.

38.10 Using the simulation that follows Example 38.7, compare the bias and the variability of the three estimators, and explain why the most precise one is the most wrong.

38.11 Compute the probability that a 500-day window's largest loss falls below the true 99.9% VaR, and explain what this means for historical simulation at that confidence.

38.12 Give the two reasons extreme value theory converts an impossible estimate into a difficult one rather than into a known answer.

38.13 Using Example 38.8, compute the long-run volatility implied by $\omega = 0.000002$, $\alpha = 0.06$, $\beta = 0.93$, and state the one-year forecast under each estimator.

38.14 Explain why EWMA is adequate for a one-day margin calculation and inadequate for a volatility term structure.

38.15 Using (38.4), recompute the VaR of Example 38.9's book at correlations of 0.30 and 0.90, and explain why the estimation error has a consistent sign.

38.16 Using Example 38.10, compute the probability that the second exposure is beyond its 99.9th percentile given that the first is, under each copula, and compare the two with the coefficient of tail dependence.

38.17 Using Example 38.11, explain why three factors keep 99% of the directional book's variance and 10% of the butterfly's, and state what a risk system must report as a result.

38.18 State the square-root-of-time rule, the assumptions it requires, and the direction of its error in a trending market.

38.19 Using Example 38.12, state each position's share of the money and share of the risk, and explain what a total-VaR limit fails to notice.

38.20 Explain how an allocation can be exact while the total it allocates is unreliable.

38.21 Using Example 38.13, state the range of exception counts consistent with a correct model at 99% over 250 days, and explain why zero exceptions is also a rejection.

38.22 Explain where the Basel zones and the Kupiec test disagree, and what the supervisor gains by setting the zones where they are.

38.23 Explain why expected shortfall cannot be backtested by counting exceptions, and how the capital framework works around this.

38.24 Using Examples 38.14 and 38.15, explain what a stress test reports that VaR does not, why a one-factor-at-a-time sensitivity report understated the scenario loss, and what the reverse stress test found about where the book's risk lies.

38.25 A book's daily loss is normal with a standard deviation of \$2 million. Using the standard-error formula of Section 38.5, compute the standard error of a 99% VaR estimated from 250, 500 and 1,000 days, each as a share of the true VaR, and state how much longer a window must be to halve it.

38.26 Repeat Example 38.5 with a dollar gamma of −\$200 million. Compute the 99% one-day loss under the delta–gamma expansion, and state by how much it exceeds the delta-only figure.

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38.27 Using the GARCH parameters of Example 38.8, show that a deviation of the variance from its long-run level halves in about 69 trading days, and compute the annualized volatility forecast for the single day 100 trading days after the variance just estimated.

38.28 Using (38.8), compute the ten-day 99% VaR of the one-day \$4.65m book of Section 38.14 at a daily autocorrelation of -0.10, and state the size and direction of the square-root rule's error.

38.29 Redo Example 38.12 at a correlation of 0.60: compute the portfolio VaR, each position's marginal, component and incremental VaR, and state which of the three sets sums to the total.

38.30 For the two-stock book of Example 38.4, find the most plausible pair of returns that loses \$1.0 million, state how many standard deviations out it lies, and compare it with the scenario in which both stocks fall by four of their own standard deviations.

FURTHER QUESTIONS

38.31 Example 38.2 constructs a subadditivity failure with two bonds. Determine the conditions on default probability, recovery, and confidence level under which the failure occurs, state how the region changes as the number of bonds grows, and explain what this implies for a real credit portfolio measured at 99%.

38.32 Example 38.6 shows the fat-tailed VaR falling below the normal VaR at 95%. Find the confidence level at which the two cross for a t distribution with ν degrees of freedom scaled to a fixed variance, determine how that crossing point moves with ν, and state what it implies for a firm that measures at 95% and fits fat tails.

38.33 Example 38.8 compares forecast term structures. Fit both estimators to a real equity index series, compare their forecasts against subsequently realized volatility at horizons from one day to one year, and determine the horizon beyond which the mean reversion earns its extra parameter.

38.34 Section 38.12 shows that a Gaussian copula has zero tail dependence. Construct a portfolio of credit exposures whose 99.9% expected shortfall is materially different under Gaussian and t copulas calibrated to the same pairwise correlations, and determine what data would be needed to choose between them.

38.35 Section 38.14 treats the horizon as a property of the position rather than of the firm. Design a liquidity-horizon scheme for a book containing listed futures, cleared swaps, and bespoke structured trades; state how you would estimate each horizon; and determine how much the aggregate measure changes against a uniform ten-day assumption.

38.36 Section 38.16 notes that expected shortfall resists backtesting. Survey the proposed tests, implement one, and determine how many observations it needs to distinguish a correct model from one miscalibrated by 20%. Then state whether that number is available in practice.

38.37 Section 38.17 solves reverse stress testing in closed form for a linear book. Take the book of Example 38.5, the sold puts on stock 1 beside the \$5 million in stock 2, find numerically the most plausible pair of returns that loses \$1.0 million under the delta–gamma expansion, compare it with the linear answer, and explain why the gamma term moves the scenario.

38.38 Section 38.18 argues that books are optimized against whatever is measured. Take a measure from this chapter, construct the portfolio that maximizes true risk subject to a fixed reported value of it, and state what supervisory rule would prevent the construction.

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